

A feasibility study on the measurement of transverse τ polarization in semitauonic B decays at the Belle II experiment.

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1 Introduction

In addition to typical observables such as decay widths or the energy spectra of decay products, semitauonic B decays allow us to measure non-trivial angular distributions and their asymmetries, owing to the large mass of the tau lepton and its two-body hadronic decay channels. At the same time, they pose a major experimental challenges: the final state contains at least two neutrinos, the visible particles have small momenta, and the extremely short lifetime of the τ lepton prevents its direct track reconstruction in the detector before decay. In semitauonic B decays, we can define three polarizations of the τ lepton:

- Longitudinal (P_L^τ) along the direction of the tau lepton's momentum.

$$P_L^\tau = \frac{\vec{s} \cdot \vec{p}_\tau}{|\vec{p}_\tau|}$$

- Transverse (P_T^τ) in the plane defined by the charm meson M and the tau lepton momentum, perpendicular to the longitudinal direction.

$$P_T^\tau = \frac{\vec{s} \cdot (\vec{p}_M \times \vec{p}_\tau)}{|\vec{p}_M \times \vec{p}_\tau|}$$

- Perpendicular ($P_\perp^\tau$) to the decay plane spanned by the charm meson M and the τ momentum.

$$P_\perp^\tau = \vec{s} \cdot \left(\frac{\vec{p}_M \times \vec{p}_\tau}{|\vec{p}_M \times \vec{p}_\tau|} \times \frac{\vec{p}_\tau}{|\vec{p}_\tau|} \right)$$

where $\vec{s}$ is the unit vector along the spin of the τ lepton, $\vec{p}_\tau$ is the momentum of the τ lepton, and $\vec{p}_M$ is the momentum of the D or D^* meson.

1.1 Transverse τ polarization

In this project, we will focus on the transverse polarization of the τ lepton, which changes sign under the time-reversal (T) operation and is equal to zero in the Standard Model (SM). Therefore, a non-zero value would unambiguously indicate the T -asymmetry. Assuming CPT conservation, this would be equivalent to CP violation. The SM describes semileptonic B decays via a single amplitude with W -boson exchange, and thus predicts no CP violation in these processes. Consequently, the observation of a non-zero transverse polarization would constitute an unambiguous signature of New Physics (NP) [2].”

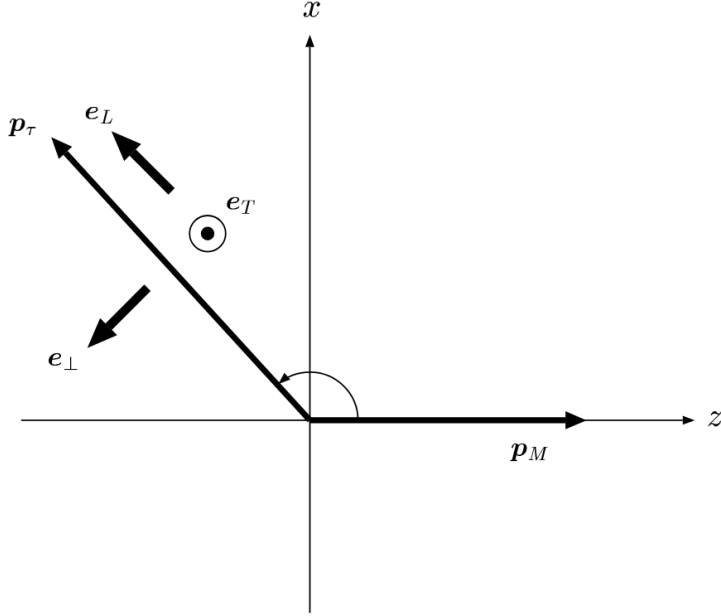


Figure 1: Kinematics and definition of the basis vectors of τ polarization.[1]

1.2 Experimental measurements

Specifically, our analysis focuses on $B^0 \rightarrow D^* \tau \nu_\tau$ decays, where $\tau \rightarrow 3\pi \nu_\tau$. The first feasibility study in B-factories on these measurements was performed by the Belle experiment [3]. It concentrated on the extensive simulation of exclusive decays $B^0 \rightarrow D^+ \tau^- \nu_\tau$ and $B^0 \rightarrow D^{*+} \tau^- \nu_\tau$, specifically utilizing the subsequent hadronic decay channel $\tau^- \rightarrow \pi^+ \pi^- \pi^- \nu_\tau$. The study demonstrated that the kinematics of these multi-neutrino final states could be successfully reconstructed by measuring both the momentum of the tag-side B^0 and the precise spatial location of the τ^- decay vertex. This method allowed for a reliable determination of the τ^- energy and direction.

We now aim to investigate the performance enhancements that can be realized in the Belle II environment, leveraging its upgraded detector sub-systems and advanced ML techniques for event reconstruction. In this project, we will address the following key points:

τ Kinematics Evaluate the energy and directional resolution of the τ lepton.

Signal Quality Assess the reconstruction quality of the triple-product observables for signal events.

Background Analyze the impact of background contributions on the measurement.

References

- [1] M. Tanaka, Charged Higgs Effects on Exclusive Semi-tauonic B Decays, [hep-ph/9411405v1](#)
- [2] G. Wu, Disentangling Non-Standard Model T-Violating Sources in Exclusive Semileptonic B Decays, [hep-ph/9706532v1](#)
- [3] M. Rozanska, K. Rybicki, On a feasibility of measuring exclusive semitauonic B0 decays at B-factories, [Acta Phys. Pol. B 29, 2065 \(1998\)](#)